

PRAMATH RAJPRASAD RAO

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PROFESSIONAL SUMMARY

Data Analyst with 3+ years in **SQL**, **Python**, **funnel analysis**, **cohort analysis**, and **A/B testing** across fintech payments and product analytics. Reduced **CAC by 18%**, improved **conversion by 14%**, and achieved **86% pipeline extraction accuracy** through experimentation, churn modeling, and **Snowflake**-based workflows. Builds automated **dashboards**, ETL pipelines, and behavioral segmentation models delivering actionable insights to cross-functional stakeholders. MS in Data Analytics Engineering with expertise spanning transaction analytics, marketplace growth, and customer **behavioral analytics**.

SKILLS

Languages & Query: Python, SQL (Presto, PostgreSQL)

Analytics & Methods: A/B Testing, Funnel Analysis, Cohort Analysis, Behavioral Analytics, Statistical Analysis, Experimentation, Churn Modeling, Data Validation, Data Visualization, Data Storytelling

Machine Learning: XGBoost, Random Forest, Classification, Clustering (BERTopic, UMAP, HDBSCAN), SHAP Interpretability, NLP, SMOTE

Data & Cloud Platforms: Snowflake, BigQuery, Airflow, ETL Pipelines, OCR, Document AI

Visualization & Reporting: Tableau, Power BI, Looker, Hex, Metabase, Excel, Matplotlib, Seaborn, Plotly

Libraries: Pandas, NumPy, Scikit-learn

Tools: Git, Jira, Jupyter Notebook

WORK EXPERIENCE

Data Analyst | Toast

Feb 2026 - Present

Customer analytics and business intelligence for Toast's restaurant platform, supporting product and operations teams across 164,000+ locations.

- Built **churn prediction** model in **Python** (XGBoost) on 14,640+ records, achieving **0.96 recall** to surface **retention** and LTV risk patterns across customer cohorts.
- Surfaced **89 behavioral segments** from 8,500+ customer records via unsupervised clustering, uncovering complaint themes absent from labeled categories.
- Executed **A/B testing** protocols with SHAP interpretability to measure **product metrics** across activation and engagement funnels, delivering **behavioral analytics** findings to product and operations teams weekly.
- Analyzed **funnel** drop-off patterns using **SQL** and **Python** across 8,500+ interaction records, translating findings into stakeholder recommendations.
- Automated reporting in **Hex** consolidating sentiment trends and churn risk scores, cutting manual prep across **3 stakeholder teams**.

Data Analytics Capstone | George Mason University

Aug 2025 - Dec 2025

Production-grade analytics pipeline built on federal contract data, applying Python, SQL, and Snowflake to enable evidence-based insights at scale.

- Built **production-grade analytics pipeline** on **Snowflake** ingesting 1,400+ federal contract PDFs using **Python**, OCR, and Document AI, achieving **86% extraction accuracy** and reducing manual review by **70%**.
- Developed **SQL**-based analytical models to surface contract-level anomalies and data quality gaps, enabling downstream **experimentation** and evidence-based insights for governance stakeholders.
- Designed scalable data ingestion, validation, and monitoring workflows ensuring continuous pipeline reliability across a large-scale, multi-format document corpus.

Data Analyst | Uber India

Jul 2022 - Jan 2024

Marketplace analytics for Uber India's City Operations and Growth teams across 100+ cities.

- Reduced **CAC by 18%** across 3 metro markets using **SQL (Presto)** and **Python** for rider cohort segmentation and campaign targeting.
- Improved trip **conversion by 14%** through **A/B testing** of geo-targeted promotions using **Python** and **Tableau**.
- Automated **ETL pipelines** in **Airflow** and **BigQuery**, cutting reporting effort by **30%** for City Operations stakeholders.
- Analyzed **funnel** drop-off patterns on 50,000+ monthly rider records using **SQL**, surfacing insights for product and growth teams.
- Maintained weekly **business intelligence** dashboards in **Tableau** tracking DAU, surge frequency, and promotional efficiency across **4 city teams**.

Data Analyst | Razorpay

Jan 2021 - Jul 2022

Transaction data analytics and merchant performance reporting for Razorpay's Risk and Operations teams.

- Queried **transaction datasets** using **SQL** to track payment gateway performance and merchant KPIs, supporting daily reporting for Sales and Operations teams.
- Maintained **Power BI** dashboards monitoring GTV trends and MoM variance across **40+ merchant accounts**, improving visibility into high-value merchant performance.
- Automated **60%+ of recurring MIS reports** using **Python** and **Excel**, reducing manual reporting effort for Risk and Operations stakeholders.
- Identified **fraud detection** signals and failed payment patterns across merchant segments using **SQL** and **Metabase**, surfacing root causes to Risk and Operations teams.

PROJECTS

Financial Fraud Detection Using Machine Learning and Behavioral Analytics

Technologies: Python (Pandas, Scikit-learn, XGBoost, Random Forest), SQL, SMOTE, Matplotlib, Seaborn, Tableau

- Built fraud detection models on 280K+ transaction records using **XGBoost** and **Random Forest**, achieving **94% ROC-AUC** through feature engineering, SMOTE oversampling, and cross-validation on imbalanced classes.
- Applied **behavioral analytics** and unsupervised clustering (Isolation Forest) to identify high-risk merchant segments with **4x higher fraud incidence**, enabling targeted risk mitigation strategies.
- Built **Tableau** dashboards to monitor fraud KPIs, feature importance, and model performance metrics across transaction types and merchant categories.

Connecticut Property Sale Prediction Using Socioeconomic Indicators

Technologies: Python (Scikit-learn, XGBoost, Pandas, Plotly), SQL, Random Forest, Statistical Analysis

- Built regression models using **XGBoost** and **Random Forest** on multi-source socioeconomic datasets across 100+ municipalities, achieving **R2 = 0.95** through feature engineering and cross-validation.
- Performed **statistical analysis** on socioeconomic indicators to identify key property value drivers, integrating 6 disparate datasets using **SQL** and **Pandas**.
- Built **Plotly** visualizations and geospatial charts communicating property value insights to non-technical stakeholders.

EDUCATION

Master of Science, Data Analytics Engineering George Mason University Fairfax, VA GPA: 3.9/4	Dec 2025
Bachelor of Technology, Computer Science PES University Bengaluru, India	Jul 2022